

FIT9132 Applied Week 4

A4-1 The Relational Model

A41.1 Relational Model Theory

- 1) **Relation** : A set of attributes consists of a head and a body,
The heading is the schema, and the body is the sets of tuples.
- 2) **Attribute** : A characteristic of an objective that we want to record
- 3) **Domain** : A set of atomic values from which attribute's value are drawn. Domain consists of a name, data type and data format. For example: cust_age: NUMBER(3)
- 4) **Tuple** : A set of related values that describe an instance of the relation (It is a row/ observation)
- 5) **Degree and Cardinality of a relation** :
 - a) **Degree** : Number of attributes
 - b) **Cardinality** : Number of Tuples
- 6) **Primary Key, Super Key and Foreign Key**
 - a) **Super Key** : A set of attributes that uniquely define a tuple.
 - b) **Primary Key** : Unique identifier (An attribute or a MINIMAL set of attributes that uniquely identifies each tuple).
 - c) **Foreign Key** : An attributes in a relation that exists in the same or another relation as Primary Key.